

Xuanyu Zhou

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Education

Zhejiang University

Urban and Rural Planning B.E. 3.93/4.0

Coursework: Principle of Urban Planning, Urban Transportation, Architecture Design, Micro/Macroeconomics

Hangzhou, Zhejiang

2021–2026 (Exp.)

University of California, Berkeley

Exchange in College of Environmental Design

Coursework: Introduce to City Planning, The Structure and Interpretation of Computer Program, Transit!

Berkeley, CA

2025

Research Experiences

MIT-UF-NEU 2025 Joint Summer Research Camp

Mentee, mentor: Dr. Xuan Jiang

Remote

Jun 2025–Oct 2025

- I participate in the development of a novel, **GPU-based computational framework designed to accurately model large-scale Urban Air Mobility (UAM) demand and optimize the operations of heterogeneous aircraft fleets.** Throughout the project, I played an integral role in the entire research lifecycle, conducting in-depth result analysis, creating data visualizations, and serving as the primary author of the research manuscript, where I synthesized our findings and formulated the final conclusions.

Monitoring-Mapping-Modeling (3M) Lab, UC Berkeley

Research Assistant, supervisor: Prof. Lu Liang

Berkeley, CA

Mar 2025–Present

- **Developing urban Sky View Factor (SVF) datasets across the United States using airborne LiDAR data,** with validation through comparative analysis of street view imagery. Responsible for processing street view data and conducting preliminary analysis. Collaborating with Ph.D. student Yuye Zhou; project ongoing.

Qizhenwenxue Undergraduate Research Program, Zhejiang University

Research Assistant, supervisor: Prof. Shuang Ma

Hangzhou, Zhejiang

Aug 2023–Present

- Authored the Introduction and Literature Review for a **case study on post-pandemic urban revival in Nagoya,** which leveraged mobile phone signaling data and a Geographic Random Forest (GRF) model. My work established the theoretical framework by synthesizing existing literature on urban mobility and resilience. This research has been published.
- Applied a GRF model to analyze a dataset of over 600,000 geotagged Weibo posts in Beijing across four months in an investigation into **the built environment's impact on mental health.** My primary responsibility was conducting the experiments and interpreting the resulting spatial patterns. The manuscript is currently under review.
- Lead the data analysis for **exploring the synergy between shared bicycles and the built environment,** processing a robust dataset of over 9 million trip records in Shenzhen using GRF. My role extended beyond generating results to include a thorough analysis of the findings and contributing to the conclusion section. A coauthored manuscript from this work is presently under review.

Academic Contributions

Articles

Ma, S., **Zhou, X.**, Cai, W., Chen, M., Li, S. (2025). Evaluating the social-economic recovery impacts of built environment for post-pandemic. *Transaction in urban data, science, and technology*.

DOI: <https://doi.org/10.1177/27541231251314132>

Ma, S., Li, W., **Zhou, X.**, Ma, T., He, Y., and Li, S. (Manuscript under review). Future life expectancy extension under bike ride, air pollution exposure, and traffic accidents through modifying the built environment and climate: An integrated modelling study. *npj Urban Sustainability*.

Li, S., Cai, W., Wu, C., Zhang, X., **Zhou, X.**, He, Y., and Ma, S. (Manuscript under review). Exploring the relationship between resident emotions and built environment in Beijing based on a geographical weighted random forest approach. *Sustainable Cities and Society*.

Zhou, X., Chen, M. (Manuscript under review). Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. *Urban Forestry & Urban Greening*.

Jiang, X., **Zhou, X.**, Zhao, Y., Cao, S., Zhao, J., Hansen, M., Sengupta, R. (Manuscript under review). From Patchwork to Network: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Advanced Air Mobility. *IEEE Transactions in Intelligent Transportation System*.

Conferences

Zhou, X., Chen, M. (2024). Greening the Gap: Examining Urban Greenery Equity in Shrinking Cities. *2024 ACSP Annual Conference (Oral Presentation)*.

Jiang, X., **Zhou, X.**, Zhao, Y., Cao, S., Zhao, J., Hansen, M., Sengupta, R. (2026). FROM PATCHWORK TO NETWORK: A Comprehensive Framework for Demand Analysis and Fleet Optimization of Urban Air Mobility. *2026 Transportation Research Board Annual Meeting (Accepted)*.

Internships

Zhejiang University Urban—Rural Planning & Design Institute Co, Ltd.	Hangzhou, Zhejiang
<i>Assistant Planner</i>	Aug 2025–Nov 2025
- Participated in the "Zhejiang Anji Economic Development Zone Application Proposal for a National-Level Zero-Carbon Industrial Park", responsible for calculating carbon emissions of the Anji Economic Development Zone.	
- Participated in the "Shangyu District Low-Altitude Economy Master Plan", the "Zhoushan City Dinghai District Low-Altitude Demonstration Pilot Application", and the "Shengzhou City Low-Altitude New Infrastructure and Application Scenarios Special Plan", contributing to partial text drafting, low-altitude infrastructure site layout and visualization, as well as research and investigation.	

Skills

Expert: Python, VSCode, ArcGIS, QGIS, Data Analysis/Visualization
Proficient: Machine Learning, R, Rhino, SketchUp, Photoshop, Illustrator, InDesign
Novice: Git/Terminal, Latex, Java, DaVinci Resolve, MATLAB, WSL

Awards

Zhejiang Provincial Government Scholarship , Zhejiang Province	2024, 2025
Second-Prize Scholarship , Zhejiang University	2023, 2024, 2025
Third-Prize Scholarship , Zhejiang University	2022
Ningbo Future Star Scholarship , Human Resources and Social Security Bureau of Ningbo	2025
Jincheng Academic Scholarship , Zhejiang University	2023
Academic Excellence Award , Zhejiang University	2022, 2023, 2024, 2025
Outstanding Student Leader , Zhejiang University	2022, 2023, 2024
Outstanding Community Service Volunteer , Zhejiang University	2025
Outstanding Member of Communist Youth League , Zhejiang University	2024
Distinguished Investigator , Innovation Center of Yangtze River Delta	2023